

Zubair Mughal

LIVE CODING CLASSES FOR KIDS, TEENS & BEGINNERS

COMPLETE PROGRAM GUIDE

The Coding & AI Pathway

A Grade 8–12 Computer Science & AI Program

Five grade-by-grade stages · twelve courses · forty-plus guided projects

12

COURSES

5

GRADE STAGES

40+

GUIDED PROJECTS

90+

WEEKS OF CURRICULUM

INTRODUCTION

A Program Built for the AI Era

Most “coding for kids” programs teach isolated tricks. This program is different: it is a single, five-stage pathway that takes a student from their first line of Python in Grade 8 to an independently designed, AI-integrated capstone application by the end of Grade 12 — with a real, portfolio-ready project at every stage along the way.

Every course is taught live, 1:1 or in small groups, and is built around one core belief: students learn to code by building things, not by watching pre-recorded lessons or memorizing syntax.

12

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40+

GUIDED PROJECTS

90+

WEEKS OF CURRICULUM

AUDIENCE

Who This Program Is For

- Students in Grade 8–12 who want a serious, structured computer-science foundation — not a scattered set of unrelated tutorials.
- Complete beginners as well as students who already know some Python or JavaScript and want to go further.
- Families who want their child to graduate high school with a real portfolio of shipped projects, not just a certificate.
- Students aiming at competitive university CS programs, internships, hackathons, or an early freelance/independent-project track.

FLEXIBLE BY DESIGN

The five-stage pathway is the recommended route, but every one of the 12 courses can also be taken as a standalone unit. Students can join at any grade level and start from wherever their current skills place them.

PROGRAM STRUCTURE

The Five-Stage Grade Roadmap

Each stage builds directly on the last — in difficulty, in the size of the projects students ship, and in how independently they're expected to work.

STAGE 1 — Grade 8

14–16 weeks

Foundations: Computational Thinking & Python

Students learn to think like programmers before they worry about syntax: breaking problems into steps, spotting patterns, and turning logic into working code using Python, the most beginner-friendly language in professional use today.

Courses: Python for Kids & Teens

Technologies: Python, VS Code, Git (intro)

Key Project: A text-based adventure game and a quiz app, built from scratch

Outcome: Reads and writes original Python programs independently; comfortable with variables, loops, conditionals, functions and lists; has shipped 4 small projects.

STAGE 2 — Grade 9

16–18 weeks

Building: Web Foundations & Intermediate Programming

Programming moves from the terminal to the browser. Students learn HTML, CSS and JavaScript to build things people can actually click on, while deepening their Python with real data structures.

Courses: JavaScript for Kids & Teens · Web Development for Kids & Teens

Technologies: HTML5, CSS3, JavaScript, Git & GitHub

Key Project: A live, deployed personal portfolio website

Outcome: Builds responsive, interactive websites end-to-end; understands the DOM, events and structured data; makes a first real commit to GitHub.

STAGE 3 — Grade 10

18–20 weeks

Applying: Full-Stack Foundations, OOP & Databases

Students add real engineering discipline: object-oriented design in Java, relational databases with SQL, and their first component-based UI framework, React — the building blocks of production software.

Courses: Java Development for Teens · SQL for Teens & Beginners · React JS for Teens & Beginners

Technologies: Java (OOP), SQL, React, Git & GitHub, REST APIs

Key Project: A database-backed inventory system and a deployed React app

Outcome: Designs classes with proper OOP principles, queries and models relational data, and builds component-based interfaces — with a GitHub portfolio to show for it.

STAGE 4 — Grade 11

20–22 weeks

Engineering: Data Structures, Backend Development & AI Foundations

The pace and rigor increase. Students learn data structures & algorithms the way real engineers do — through guided practice, not memorization — build their own backend APIs with Node.js, and get a responsible, practical introduction to generative AI.

Courses: DSA for Teens · Node.js for Teens & Beginners · Generative AI for Teens

Technologies: Data Structures & Algorithms, Node.js, Express, Generative AI APIs

Key Project: A full-stack CRUD app with a self-built backend, plus an AI-powered study helper

Outcome: Analyzes algorithmic efficiency, ships and deploys backend APIs, and builds AI-powered features responsibly and critically.

STAGE 5 — Grade 12

22–24 weeks (incl. capstone)

Specializing: Advanced AI, Data Science, System Design & Capstone

The pathway culminates in specialization and independence. Students work with real datasets, design autonomous AI agents, reason about system architecture at scale, and plan, build, and ship an original capstone application end-to-end — the centerpiece of their portfolio.

Courses: Data Science for Teens · Agentic AI for Teens · System Design for Teens · Capstone Project

Technologies: pandas & data viz, AI agents & tool use, System architecture, Full-stack + AI

Key Project: An original, independently designed and shipped capstone application

Outcome: Leaves the program with a 15+ project portfolio, a capstone app they can demo in an interview, and college/career-ready technical and communication skills.

THE FULL STACK

Technology & Skills Covered

Across the pathway, students are exposed to a deliberately broad, modern stack — sequenced so nothing is introduced before a student is ready for it.

LANGUAGES

Python

JavaScript (ES6+)

Java

SQL

HTML5 / CSS3

WEB & FRONTEND

React

DOM APIs

Responsive Design

Git & GitHub

BACKEND & DATA

Node.js

Express

REST APIs

Relational Databases

AI & MODERN TECH

Generative AI

Prompt Engineering

AI Agents & Tool Use

Responsible AI

ENGINEERING PRACTICE

Data Structures & Algorithms

System Design

Testing & Debugging

Deployment

COURSE CATALOG

All 12 Courses at a Glance

#	COURSE	GRADE	LEVEL	DURATION
01	Python for Kids & Teens	Grade 8–12	Beginner	14–16 weeks
02	JavaScript for Kids & Teens	Grade 8–12	Beginner	12 weeks
03	Web Development for Kids & Teens	Grade 8–12	Beginner	14 weeks
04	Data Science for Teens	Grade 9–12	Intermediate	14 weeks
05	Agentic AI for Teens	Grade 9–12	Intermediate–Advanced	10 weeks
06	Generative AI for Teens	Grade 8–12	Beginner–Intermediate	8 weeks
07	Java Development for Teens	Grade 9–12	Intermediate	16 weeks
08	React JS for Teens & Beginners	Grade 9–12	Beginner–Intermediate	10 weeks

#	COURSE	GRADE	LEVEL	DURATION
09	Node.js for Teens & Beginners	Grade 9–12	Beginner–Intermediate	10 weeks
10	SQL for Teens & Beginners	Grade 9–12	Beginner	8 weeks
11	DSA for Teens	Grade 9–12	Intermediate–Advanced	16 weeks
12	System Design for Teens	Grade 10–12	Advanced	10 weeks

Full, module-by-module curriculum guides for every individual course are available as separate downloads alongside this guide.

LEARNING BY BUILDING

Project Progression

Every stage adds a new layer of project complexity. By the end of the program, a student's portfolio traces a clear, visible arc — exactly what admissions committees and interviewers want to see.

GRADE	PROJECT TYPE	EXAMPLE
Grade 8	Guided beginner mini-projects	Guessing games, calculators, a text adventure
Grade 9	Interactive & visual projects	A deployed personal portfolio website
Grade 10	Applied, database-backed projects	An OOP inventory system, a deployed React app
Grade 11	Full-stack & AI-powered projects	A self-built backend API, an AI study-helper agent
Grade 12	Independent capstone project	An original, end-to-end AI-integrated application

FUTURE-READY

Why This Matters in the AI Era

AI tools can now generate code — which makes the underlying skills more valuable, not less. Students who understand how software actually works can direct, evaluate, and debug AI-assisted work; students who don't are left unable to tell good output from broken output. This is why the pathway pairs traditional computer-science fundamentals — logic, data structures, systems thinking — with hands-on, responsible exposure to generative and agentic AI, rather than treating AI as a separate add-on unit.

GRADE 12

Capstone & Portfolio Building

The program closes with an independently scoped capstone project: the student picks a real problem, designs the system, builds the full-stack (and where relevant, AI-integrated) application, deploys it, and documents it — with guided support, not a fixed template to copy.

- Scoping — defining a problem worth solving and a realistic project plan
- Design — applying system-design thinking from prior coursework
- Build — implementing the full application, frontend to backend (and AI, where relevant)
- Ship — deploying the project live, not just running it locally
- Present — documenting and demoing the project as they would in a real interview

BY GRADUATION, STUDENTS HAVE

A portfolio of 15+ real, working projects across five stages of increasing complexity, a public GitHub history, and a capstone application they can confidently demo and discuss in depth — the combination that college admissions readers, hackathon judges and internship interviewers actually look for.

BEYOND THE CODE

College & Career Readiness

- A demonstrable, project-based portfolio to support university CS and engineering applications
- Direct exposure to the languages and concepts used in most intro college CS courses (Python, Java, DSA)
- Practice explaining technical decisions clearly — a skill assessed in interviews at every level
- A foundation for competitive programming, hackathons, and early internship or freelance work
- Genuine, hands-on AI literacy — increasingly expected in technical fields far beyond software

NEXT STEPS

Getting Started

Every student starts from a different place, so the first step is always a conversation — not a fixed enrollment form. Tell us the student's grade, current experience and interests, and we'll recommend the right starting course from the pathway.

MODULAR ENROLLMENT

Families can enroll in the full multi-year pathway, or in any single course on its own — both 1:1 and small-group live formats are available. Every course listed in this guide is also available as its own detailed curriculum download.

This document describes an original curriculum designed by Zubair Mughal for live, project-based coding instruction. It is not affiliated with, and does not reproduce material from, any other coding-education provider.